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De Morgan's Laws

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SUBJECT
Mathematics Foundation T

SEMESTER
Semester 1

// Notes Summary

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title: "De Morgan's Laws" semester: subject: "Mathematics Foundation to Computer Science I" subjectSlug: "mathfoundation1" unit: "Unit I: Set, Relation and Function" unitSlug: "unit1setrelationfunction" order: description: "Complement duality laws, transformation of unions

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Topics Covered

Intermediate
Difficulty Level

Why this topic exists

De Morgan's Laws explain how the complement operation interacts with unions and intersections. They allow you to transform complex set expressions into simpler, equivalent forms. In computer science, these laws are essential for simplifying set-based search queries, optimizing database operations, and refining conditional execution paths.

Statement of De Morgan's Laws

✓ [VALID / RULE] Duality Rule

Distribute the complement to each set inside the parentheses and flip the operational symbol ($\cup \iff \cap$).

1. First Law: Complement of a Union

✦ [DEFINITION] First Law

The complement of the union of two sets equals the intersection of their individual complements:

$$(A \cup B)' = A' \cap B'$$

- **In Words:** "An element not in A or B is neither in A nor in B." - **Example:** Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$. - **Complements:** $A' = U - A = \{3, 4, 5\}$, $B' = U - B = \{1, 4, 5\}$ - **Left-Hand Side (LHS):** $A \cup B = \{1, 2, 3\} \implies (A \cup B)' = \{4, 5\}$ - **Right-Hand Side (RHS):** $A' \cap B' = \{3, 4, 5\} \cap \{1, 4, 5\} = \{4, 5\}$ - **Result:** $\text{LHS} = \text{RHS} = \{4, 5\}$
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2. Second Law: Complement of an Intersection

✦ [DEFINITION] Second Law

The complement of the intersection of two sets equals the union of their individual complements:

$$(A \cap B)' = A' \cup B'$$

- **In Words:** "An element not in both A and B is outside A, outside B, or outside both." - **Example:** Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$. - **Complements:** $A' = U - A = \{3, 4, 5\}$, $B' = U - B = \{1, 4, 5\}$ - **Left-Hand Side (LHS):** $A \cap B = \{2\} \implies (A \cap B)' = \{1, 3, 4, 5\}$ - **Right-Hand Side (RHS):** $A' \cup B' = \{3, 4, 5\} \cup \{1, 4, 5\} = \{1, 3, 4, 5\}$ - **Result:** $\text{LHS} = \text{RHS} = \{1, 3, 4, 5\}$